

Genetics genius

Gregor bred pairs of parent plants with different characteristics, such as purple or white flowers. When the offspring plants grew, he recorded whether each one had purple or white flowers. Learning physics had taught Gregor to repeat experiments again and again, so next he bred pairs of the offspring plants.

Over time, Gregor collected a huge amount of data and was able to see patterns that no one else had noticed before. He determined that some plants must carry information for **both** purple **and** white flowers, but could only pass on one of these to their offspring. He created rules to describe how these simple “hereditary factors” worked. In 1909, these factors were renamed “genes” and the science of genetics began.

In the 20th century, Mendel's work at the monastery helped other scientists to show how variation—the natural range of differences within a species—came about and was passed on.

Strong genes

When Gregor paired plants with purple flowers and plants with white flowers, the first generation of their offspring all had purple flowers.



Gene pairs

When Gregor bred new plants from the first generation offspring, plants with white flowers reappeared! The ratio was one white flower for every three purple ones.

			
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